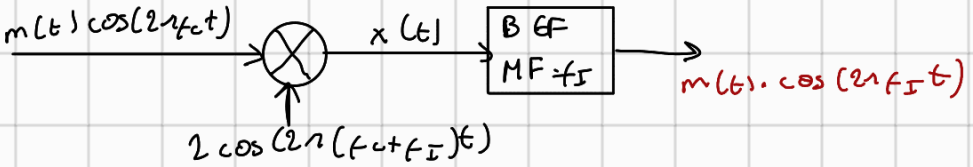
Örnek: Frekans karıştırıcı (Frequency mixer)

Changes carrier frequency f_c to f_I (f_I: düşük, f_c: yüksek)

i.e. $m(t)\cos(2\pi f_c t) \rightarrow m(t)\cos(2\pi f_I t)$



$$x(t) = 2m(t)\cos(2\pi f_c t)\cos[2\pi(f_c \pm f_I)t]$$

$f_k = f_c - f_I$ veya $f_k = f_c + f_I$

$$= m(t)[\cos(2\pi(f_c - f_k)t) + \cos(2\pi(f_c + f_k)t)]$$

$f_k = f_c - f_I \Rightarrow x(t) = m(t)[\cos(2\pi f_I t) + \cos(2\pi(2f_c - f_I)t)]$

$f_k = f_c + f_I \Rightarrow x(t) = m(t)[\cos(2\pi f_I t) + \cos(2\pi(2f_c + f_I)t)]$

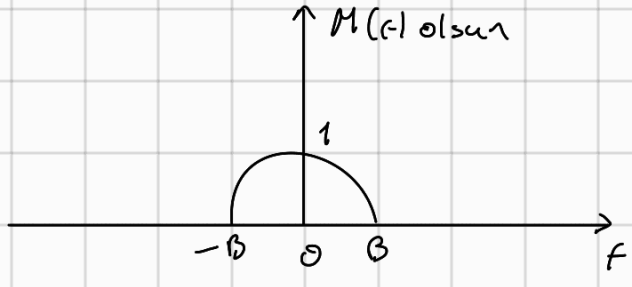
Bant geçiren filtreden geçerler

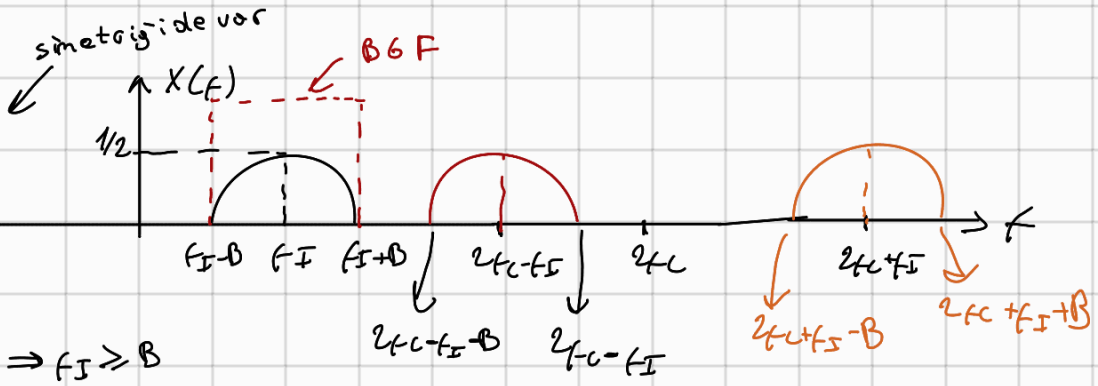
BGF'den geçmez

$f_I = 450\,kHz$

$f_c = 1000\,kHz$

Çok yüksek frekanslarda devre tasarlamak zor o yüzden düşürüyoruz





★ $f_I - B \geq 0 \Rightarrow f_I \geq B$

Saf olarak elde edebilmek için

$2f_c - f_I - B \geq f_I + B \Rightarrow 2(f_c - f_I) \geq 2B \Rightarrow (f_c - f_I) \geq B$

4.2.4 kağıttaki soruları çözelim

4.2-4 You are asked to design a DSB-SC modulator to generate a modulated signal $km(t)\cos\omega_c t$, where $m(t)$ is a signal band-limited to B Hz. Figure P4.2-4 shows a DSB-SC modulator available in the stock room. The carrier generator available generates not $\cos\omega_c t$, but $\cos^3\omega_c t$. Explain whether you would be able to generate the desired signal using only this equipment. You may use any kind of filter you like.

(a) What kind of filter is required in Fig. P4.2-4?

(b) Determine the signal spectra at points b and c, and indicate the frequency bands occupied by these spectra.

(c) What is the minimum usable value of ω_c?

(d) Would this scheme work if the carrier generator output were $\cos^2\omega_c t$? Explain.

(e) Would this scheme work if the carrier generator output were $\cos^n\omega_c t$ for any integer $n \geq 2$?

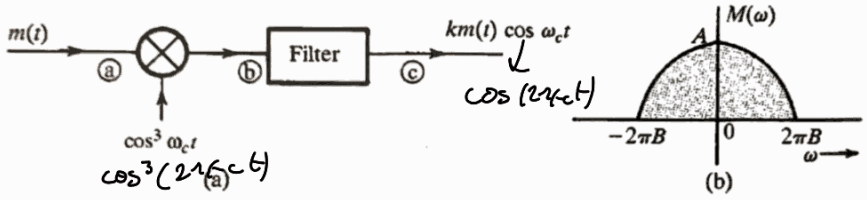
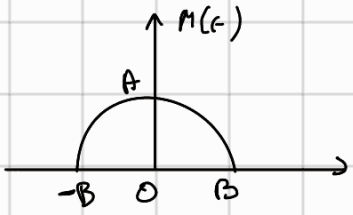


Figure P4.2-4

DSB-SC-Am modulator

BG{m(t)} = B Hz



a) $x(t) = m(t)\cos^3(2\pi f_c t)$

$\cos^3(\alpha) = \cos(\alpha)\cdot\cos^2(\alpha)$

$= \cos\alpha \ \frac{1}{2}[1+\cos(2\alpha)]$

$= \frac{1}{2}\cos(\alpha) + \frac{1}{2}\cos(\alpha)\cos(2\alpha)$

$= \frac{1}{2}\cos(\alpha) + \frac{1}{4}[\cos(\alpha)+\cos(3\alpha)]$

$\cos^3\alpha = \frac{3}{4}\cos(\alpha) + \frac{1}{4}\cos(3\alpha)$

$x(t) = m(t)\cdot\cos^3(2\pi f_c t)$

$= \frac{3}{4}m(t)\cos(2\pi f_c t) + \frac{1}{4}m(t)\cos(2\pi 3 f_c t)$

($2\pi f_c t = \alpha$)

→ 2B bant genişliğine sahip

Bu işareti geçirmeli

Bu zaman filtreniz

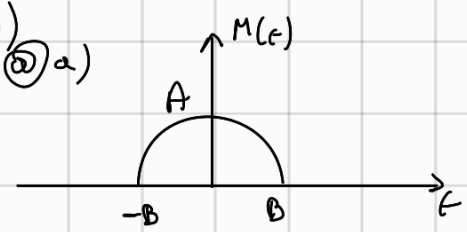
Bzaman BGF

MF = fc

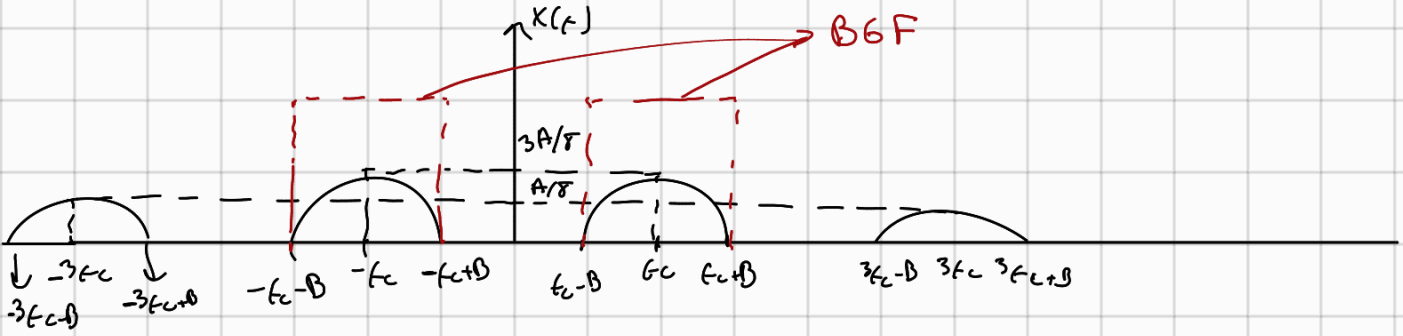
BG = 2B

b)

@a)



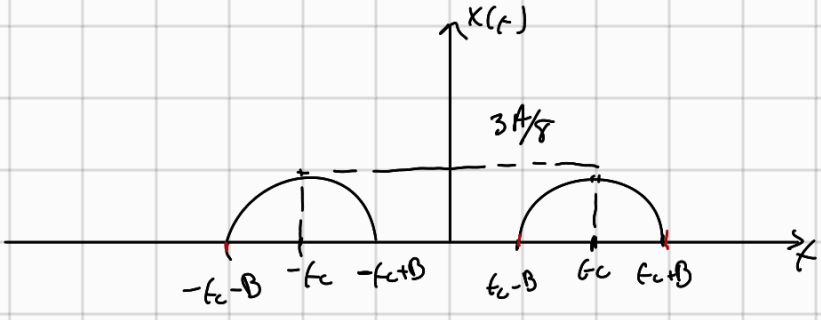
@b $$X(f) = \frac{3}{8}\left[M(f-f_c)+M(f+f_c)\right] + \frac{1}{8}\left[M(f-3f_c)+M(f+3f_c)\right]$$



@c

$$y(t) = \frac{3}{4} m(t) \cos(2\pi f_c t)$$

$$Y(f) = \frac{3}{8}\left[M(f-f_c)+M(f+f_c)\right]$$



c) $f_c - B \geq 0 \Rightarrow \boxed{f_c \geq B}$

$3f_c - B \geq f_c + B \Rightarrow 2f_c \geq 2B$

$\Rightarrow f_c \geq B$

d) $\cos^2(2\pi f_c t) = \frac{1}{2}\left[1+\cos(2\pi 2 f_c t)\right]$

$x(t) = m(t)\cos^2(2\pi f_c t)$

$= \frac{1}{2} m(t) + \frac{1}{2} m(t)\cos(2\pi 2 f_c t)$

Hayır.

e) $\cos^n(2\pi f_c t)$, $n \geq 2$

Even, n tek ise

Problem 4-2-9, lathi

4.2-9 System shown in Fig. P4.2-9 is used for scrambling audio signals. The output $y(t)$ is the scrambled version of the input $m(t)$.

(a) Find the spectrum of the scrambled signal $y(t)$.

(b) Suggest a method of descrambling $y(t)$ to obtain $m(t)$.

A slightly modified version of this scrambler was first used commercially on the 25-mile radio-telephone circuit connecting Los Angeles and Santa Catalina island.

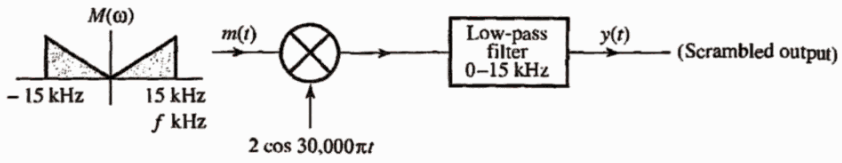
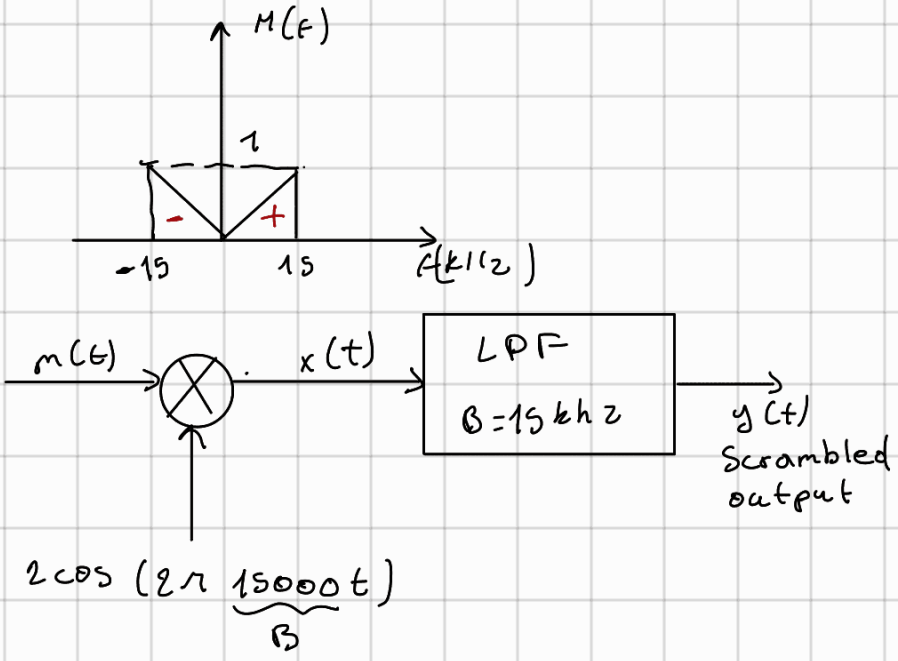
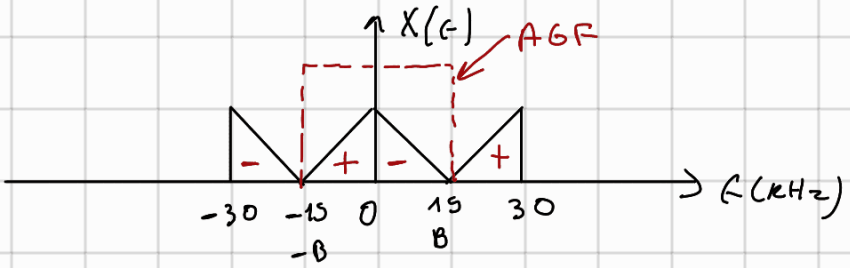


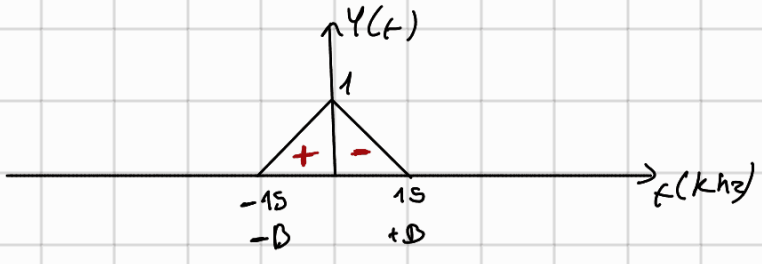
Figure P4.2-9



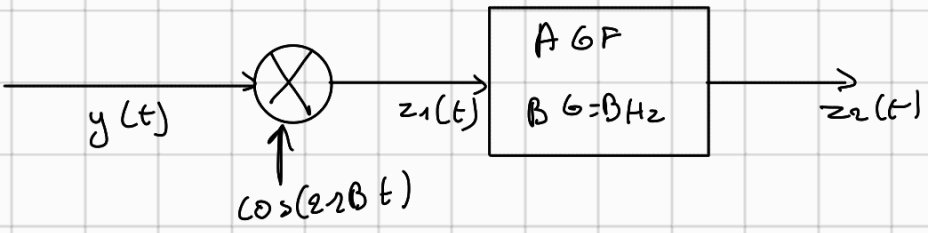
a) $x(t) = 2\, m(t) \cos(2\pi B t)$

$X(f) = M(f-B) + M(f+B)$

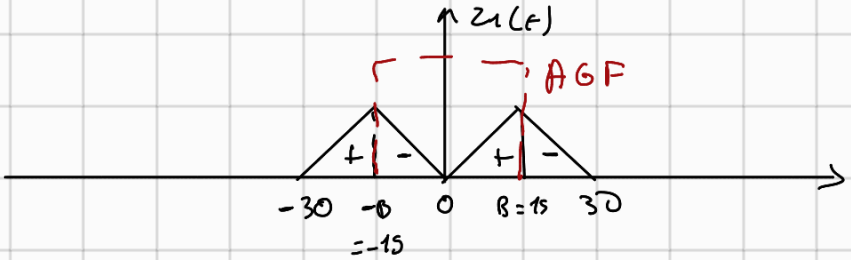


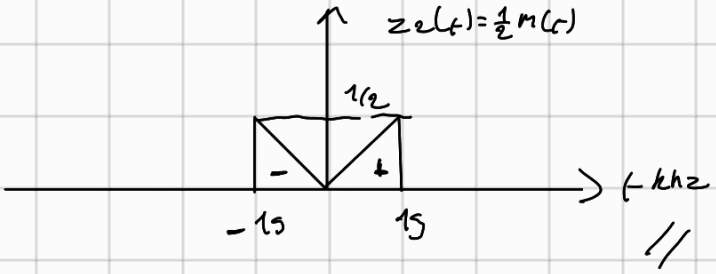


b) Descrambiling : $y(t) \rightarrow m(t)$



$$z_1(t) = y(t)\cos(2\pi Bt) \rightarrow Z_1(f) = \frac{1}{2}[Y(f-B) + Y(f+B)]$$





//

Bu soruya niye gaptık ??

Mesaj işaretini şifrelemek için mesaj işareti karıştırılır. Böylece mesajı okumak için deşifre etmek lazım.

DSB - Large carrier (LC) - AM

Büyük taşıyıcılı (BT) - GYB - GM

Conventional Am / Commercial Am ≡ Am
(Ticari Gmj)

faz tahminsiz demodülasyon gapılabilir.
Burda maksat alıcıyı basitleştirmek. Alıcı daha ucuz olur.

İlk sayısal haberleşme s. telegraf, analog ta iseradyo.

Verici karmaşık olur alıcı ise ucuz olur.

$$y(t) = A_c[1 + \mu m_n(t)] \cos(2\pi f_c t)$$

modüle eden işaret

$0 < \mu < 1$ modülasyon indeksi

$m_p = \text{maks}|m(t)| \Rightarrow m_n(t) = \frac{m(t)}{m_p}$

(m_p: peak value; $m_n(t)$: normallestirilmiş mesaj işareti)

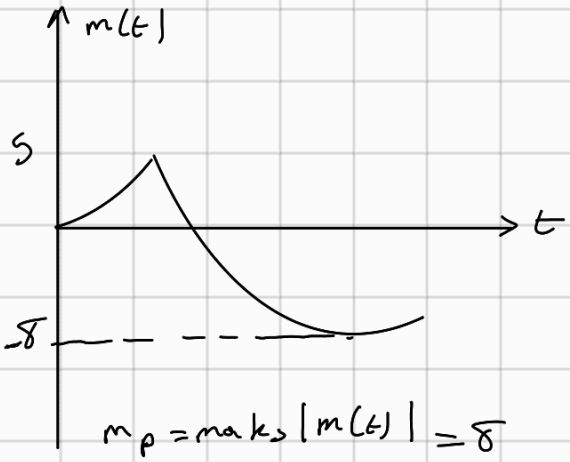
bütün t'ler için $\forall t \quad |m_n(t)| \leq 1$

$$-1 \leq m_n(t) \leq 1$$

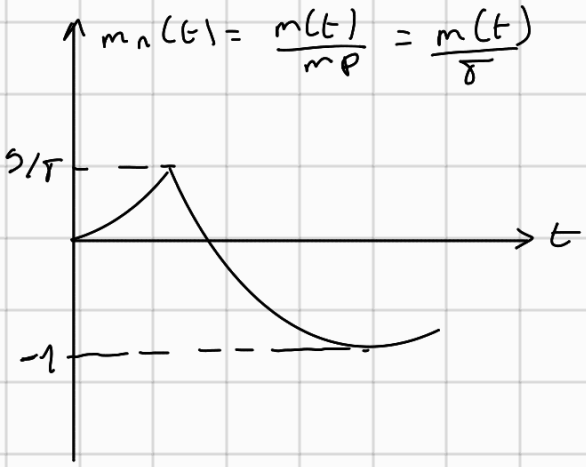
$\forall t \quad -1 \leq m_n(t) \leq 1$

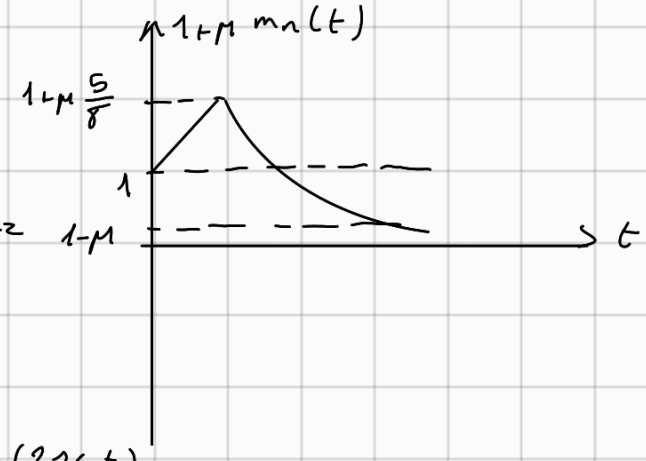
$\Rightarrow \forall t \quad 0 < 1 + \mu m_n(t) < 2$

$\Rightarrow \forall t \quad 1 + \mu m_n(t) \geq 0$ modüle eden işaret her zaman pozitif



$m_p = \text{maks}|m(t)| = \delta$

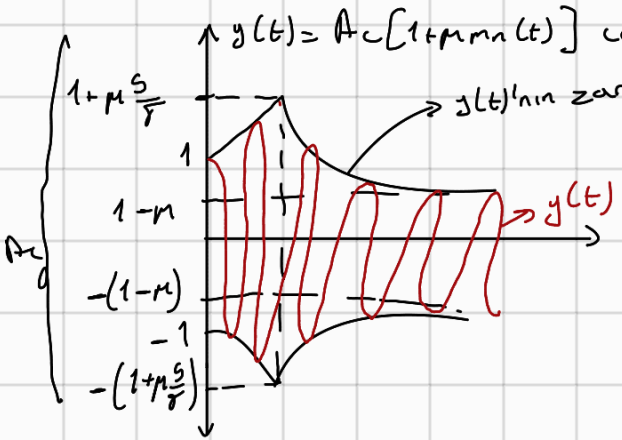




μ 1 olmasını tercih etmiyoruz

$0 < \mu < 1$ arasında seçilirse

$\Rightarrow \forall t \quad 1 + \mu m_n(t) > 0$

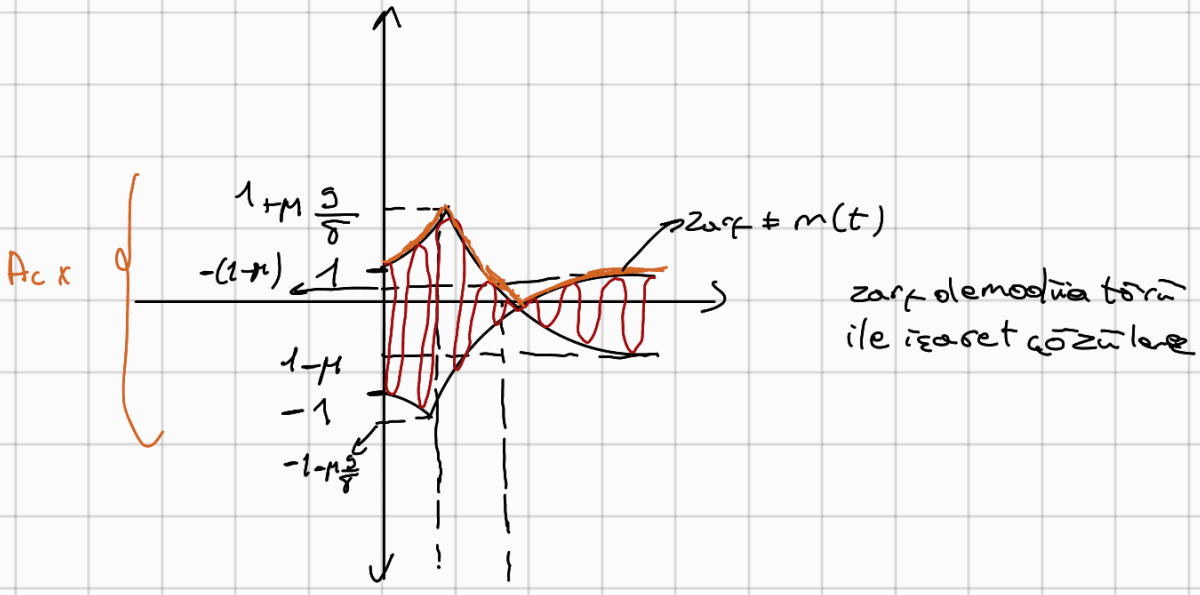


mesaj işareti zarfta gizli

Faz tahminsiz demodülasyon

Zarf demodülatörü (Envelope detector)

Eğer $\mu > 1$

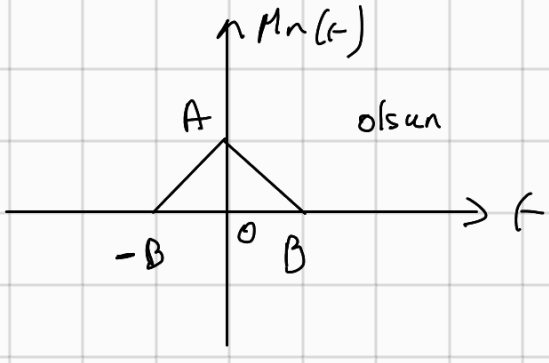


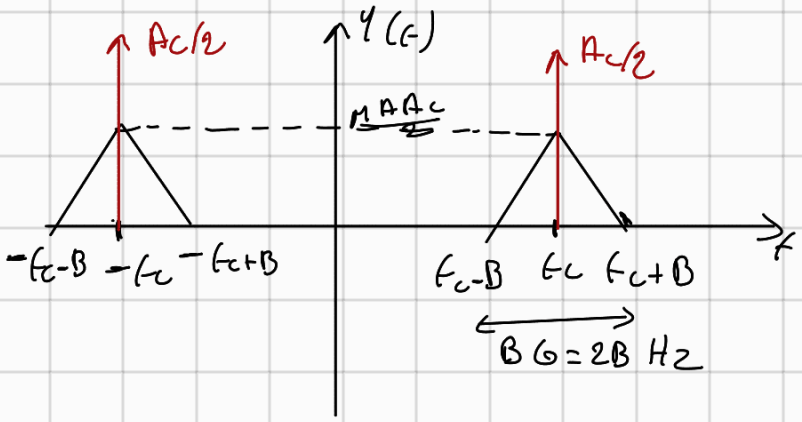
★ Eğer $\mu > 1$ (Aşırı modülasyon)
Overmodulation

$$y(t) = A_c[1+\mu m_n(t)]\cos(2\pi f_c t)$$
$$= A_c\cos(2\pi f_c t) + \mu A_c m_n(t)\cos(2\pi f_c t)$$

BT (Büyük Taşıyıcı) — TB-GYB-GM

$$Y(f) = \frac{A_c}{2}[\delta(f-f_c)+\delta(f+f_c)] + \frac{\mu A_c}{2}[M_n(f-f_c)+M_n(f+f_c)]$$





Örnek: Ton modülasyonu $\Rightarrow m(t) = c\cos(2\pi f_m t)$, $f_m \ll f_c$

for $\mu = \frac{1}{2}$, sketch DSB-LC-AM signal $g(t)$

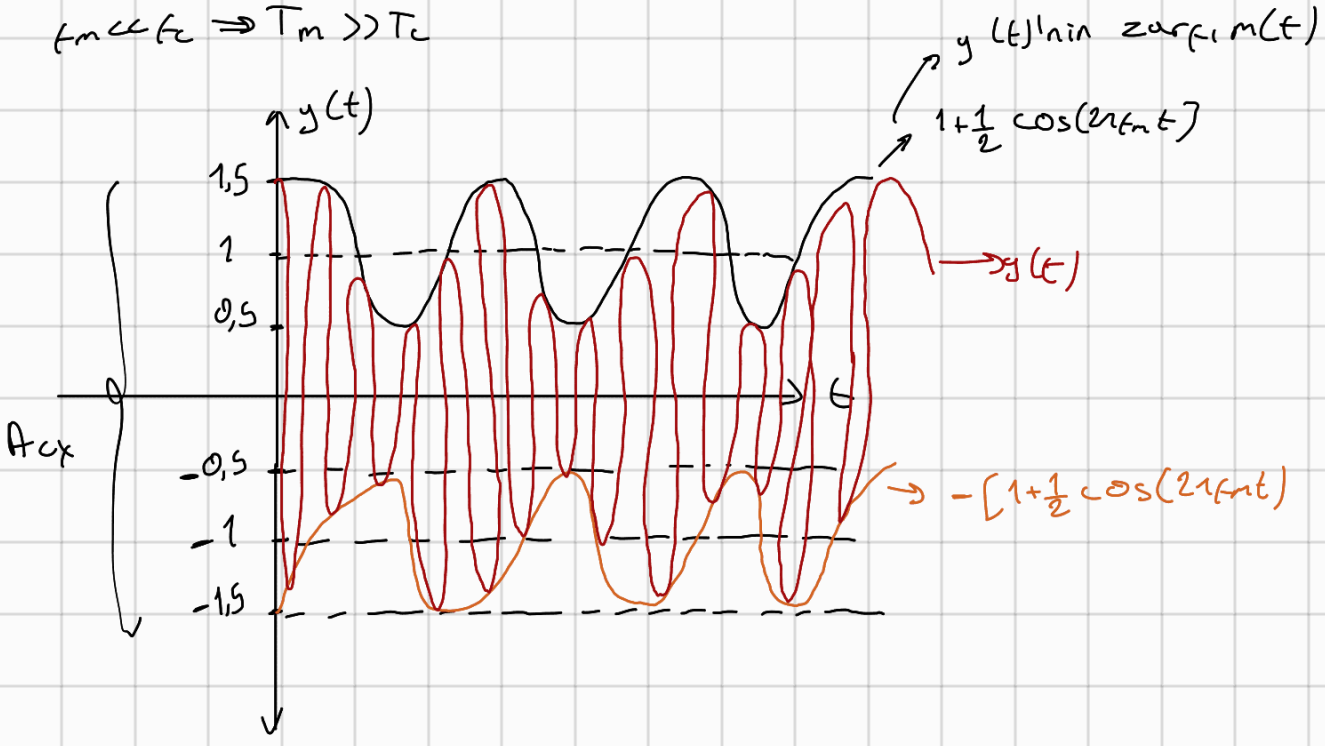
// $Y(f)$

$m(t) = c\cos(2\pi f_m t) \Rightarrow m_p = c \Rightarrow m_n(t) = \frac{m(t)}{m_p} = \cos(2\pi f_m t)$

$c > 0$

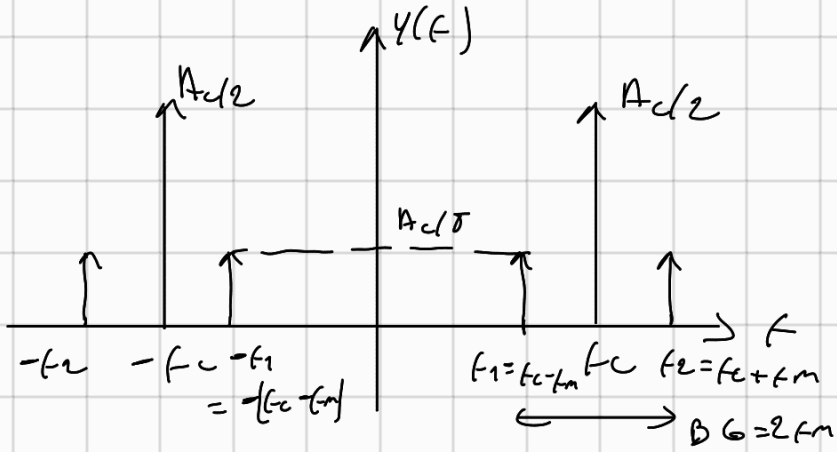
$$g(t) = A_c[1 + \mu m_n(t)]\cos(2\pi f_c t)$$
$$= A_c\left[1 + \frac{1}{2}\cos(2\pi f_m t)\right]\cos(2\pi f_c t)$$

$f_m \ll f_c \Rightarrow T_m \gg T_c$



$$g(t) = A_c\left[1 + \frac{1}{2}\cos(2\pi f_m t)\right]\cos(2\pi f_c t)$$
$$= A_c\cos(2\pi f_c t) + \frac{A_c}{2}\cos(2\pi f_m t)\cos(2\pi f_c t)$$
$$= A_c\cos(2\pi f_c t) + \frac{A_c}{4}\left[\cos(2\pi(f_c - f_m)t) + \cos(2\pi(f_c + f_m)t)\right]$$

$$Y(f) = \frac{A_c}{2}\left[\delta(f - f_c) + \delta(f + f_c)\right] + \frac{A_c}{8}\left[\delta(f - f_1) + \delta(f + f_1) + \delta(f - f_2) + \delta(f + f_2)\right]$$



BT - GYB - GM'nin Gücü

$$y(t) = A_c[1+\mu m_n(t)]\cos(2\pi f_c t)$$

$$= \underbrace{A_c \cos(2\pi f_c t)}_{\text{Taşıyıcı (carrier)}} + \underbrace{\mu A_c\, m_n(t) \cos(2\pi f_c t)}_{\text{yan bandlar (sidebands)}}$$

Taşıyıcının gücü $= P_c = \frac{A_c^2}{2}$

Yan bandların gücü $= P_s = \mu^2 A_c^2 \cdot P_{m_n}$

$$P_{m_n} = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} |m_n(t)|^2 dt$$

$$P_y = P_c + P_s = \frac{A_c^2}{2} + \frac{\mu^2 A_c^2}{2} P_{mn}$$

$$\text{Verim} = V = \frac{P_s}{P_y} = \frac{P_s}{P_c + P_s} = \frac{\mu^2 \frac{A_c^2}{2} P_{mn}}{\frac{A_c^2}{2} + \mu^2 \frac{A_c^2}{2} P_{mn}}$$

(Efficiency)

$$V = \frac{\mu^2 P_{mn}}{1+\mu^2 P_{mn}}$$

Ton modülasyonu $\rightarrow$ $m(t) = c\cos(2\pi f_m t) \rightarrow m_n(t) = \cos(2\pi f_m t)$

$$P_{m_n} = \frac{1}{2}$$

$$\Rightarrow V = \frac{\mu^2 \cdot \frac{1}{2}}{1+\mu^2 \frac{1}{2}} = \frac{\mu^2}{2+\mu^2}$$

ton modülasyonu verimi $V = \frac{\mu^2}{2+\mu^2}$

$\mu = 1 \rightarrow V_{max} = \frac{1}{3} = \%33$

Pratikte verim $V \simeq \%25$ olur

Neden yapıyoruz alıcı ucuzluyor verici pahalı oluyor, böyle herkes alabiliyor alıcıyı.

Verim sorabilir

* m_n'in ortalaması pratikte 0 o yüzden $\frac{1}{2} m_n(t)$ nin gücü yaklaşık 0 o yüzden iki işaretin çarpımından gelen katkı $(a+b)^2 = a^2+b^2+2ab$ görmezden gelinir.

Örnek: $m(t) = 5\cos(2\pi f_m t)$ $\mu = \frac{1}{2}$, $A_c = 10$

$y(t) = ?$ $P_c = ?$ $P_s = ?$ $V = ?$

$m_p = 5 \Rightarrow m_n(t) = \cos(2\pi f_m t)$

$y(t) = A_c[1 + \mu m_n(t)] \cos(2\pi f_c t)$

$= 10[1 + \frac{1}{2}\cos(2\pi f_m t)] \cos(2\pi f_c t)$

$P_c = \frac{A_c^2}{2} = \frac{10^2}{2} = 50\,W$

$P_s = \frac{\mu^2 A_c^2}{2} \cdot P_{mn} = \frac{(\frac{1}{2})^2 \cdot 10^2}{2} = \frac{1}{2} = \frac{100}{10} = 6,25\,W$

$P_y = P_c + P_s = 56,25\,W$

$V = \frac{P_s}{P_c + P_s} = \frac{6,25}{50+6,25} = \frac{6,25}{56,25} = \frac{1}{9}$ veya $V = \frac{\mu^2}{2+\mu^2} = \frac{\frac{1}{4}}{2+\frac{1}{4}} = \frac{1}{9}$

Örnek (Proakis)

$m(t) = 3\cos(200\pi t) + \sin(600\pi t)$ is used to modulate the carrier $c(t) = \cos(2\pi \cdot 10^5 t)$. The modulation index is $\mu = 0,85$. Determine the power in the carrier component and in the sideband components of the modulated signal.

$m(t) = 3\cos(200\pi t) + \sin(600\pi t)$

Max değeri bulmak için türev alıp 0'a eşitleriz

$m'(t) = -3 \cdot 200\pi \sin(200\pi t) + 600\pi \cos(600\pi t) = 0$

$600\pi \sin(200\pi t) = 600\pi \cos(600\pi t)$

$\sin(200\pi t) = \cos(600\pi t)$

$\cos(\frac{\pi}{2} - 200\pi t) = \cos(600\pi t)$

$\sin\alpha = \cos(\frac{\pi}{2} - \alpha)$

$\Rightarrow \frac{\pi}{2} - 200\pi t = 600\pi t$

$\frac{\pi}{2} = 800\pi t \Rightarrow t = \frac{1}{1600}$

$m(t)\Big|_{t=\frac{1}{1600}} = m(\frac{1}{1600}) = 3,6955 = m_p$

$m_n(t) = \frac{m(t)}{3,6955} = 0,8118\cos(200\pi t) + 0,2706\sin(600\pi t)$

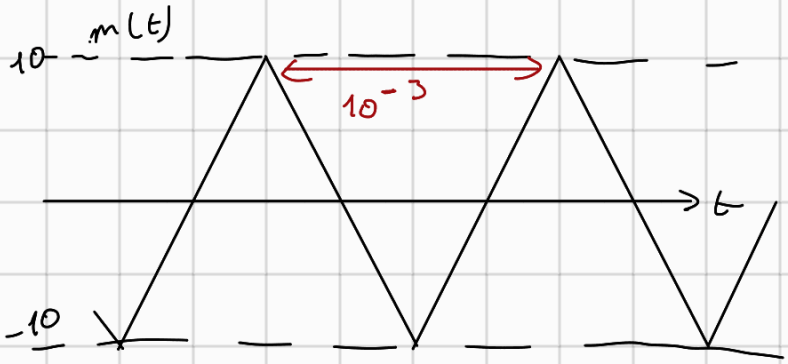
$P_{mn} = \frac{1}{2}[0,8118^2 + 0,2706^2] = 0,3661\,W$

$P_c = \frac{1}{2}$

$P_s = \frac{\mu^2 A_c^2}{2} P_{mn} = \frac{(0,75)^2 (1)^2}{2} 0,3661 = 0,1323w$

* yaklaşık olarak maksimum bulunabilir
böyle cos tolar veya sin 1 seçilebilir

Örnek Lathi, Pr. 4.3-3

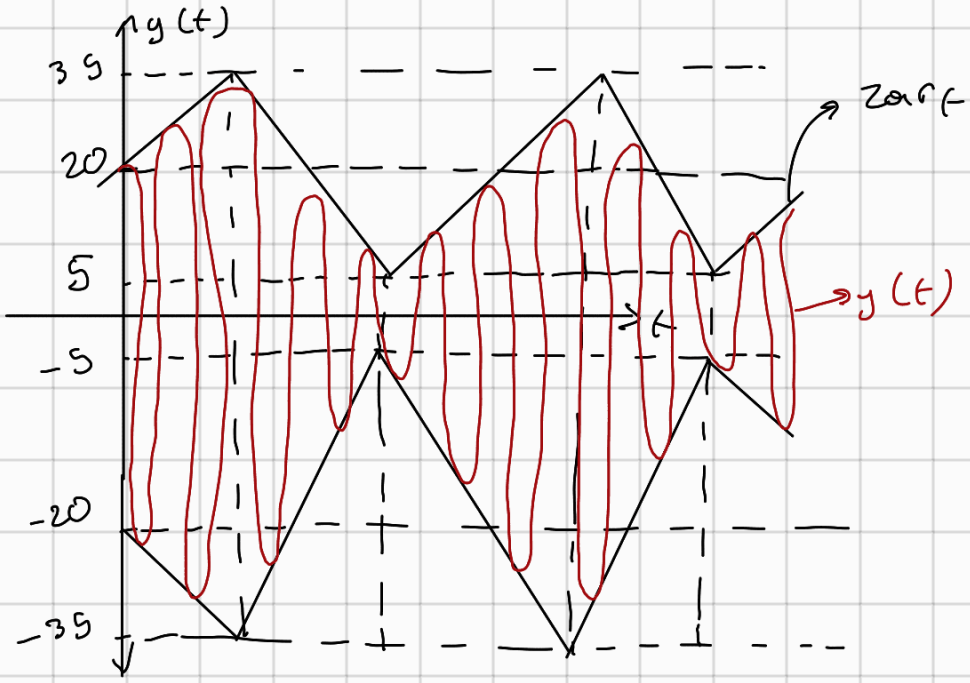


$A_c = 20$, $\mu = 0,26$

a) Sketch $y(t) = ?$

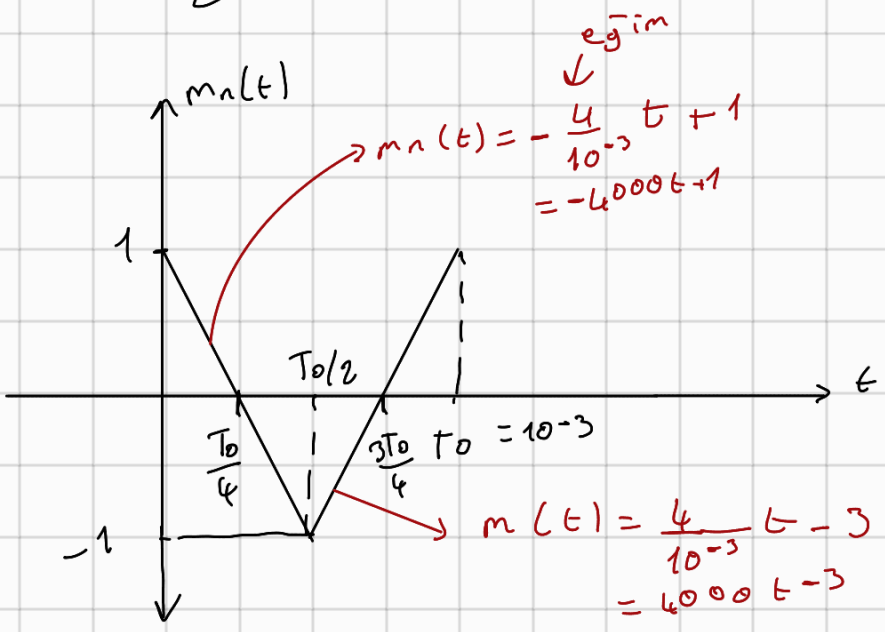
b) $P_c, P_s, P_y, V = ?$

$$y(t) = A_c [1 + \mu m_n(t)] \cos(2\pi f_c t)$$
$$= 20 [1 + 0,75\, m_n(t)] \cos(2\pi f_c t)$$



b) $P_c = \frac{A_c^2}{2} = \frac{20^2}{2} = 200\,W$

$P_s = \frac{\mu^2 A_c^2}{2} P_{mn}$



$$P_{mn} = \frac{1}{T_0} \int_{T_0} |m_n(t)|^2 dt$$

$$= \frac{1}{10^{-3}} \left[\int_0^{10^{-3}/2} (-4000t + 1)^2 dt + \int_{10^{-3}/2}^{10^{-3}} (4000t - 3)^2 dt \right]$$

$$= 0,3334\,W$$

$$P_s = \frac{\mu^2 A_c^2}{2} P_{mn} = \frac{(0,75)^2 \times 20^2}{2} \times 0,3334 = 37,5\,W$$

$$P_y = P_c + P_s = 200 + 37,5 = 237,5\,W$$

$$\nu = \frac{P_s}{P_y} = \frac{37,5}{237,5} = 0,1578 \approx 0,16 = \%16$$